

IMPLEMENTATION OF DATA HIDING METHODS FOR RESOURCE CONSTRAINED IoT DEVICES

NISHMANUL HAQ S (9517202203103)

SHAH MUHAMMED ILYAS M (9517202203141)

Dr. N. Sivasankari M.E.,
Ph.D., Internal guide,
Associate Professor,
Department of Electronics and
Communication Engineering,
Mepco Schlenk Engineering
College, Sivakasi.

Closure Report

The final year project titled "Implementation of Data Hiding Methods for Resource-Constrained IoT Devices" focused on developing a secure, efficient, and lightweight encryption framework suitable for modern Internet of Things (IoT) environments. With the rapid growth of IoT devices across industrial, domestic, and smart city applications, ensuring data security has become a critical challenge. Traditional encryption techniques such as AES and RSA, although highly secure, are not well-suited for resource-constrained microcontrollers due to their high computational and memory requirements. Moreover, existing encryption systems do not address the problem of securely displaying data on embedded devices, where sensitive information may be exposed visually on screens.

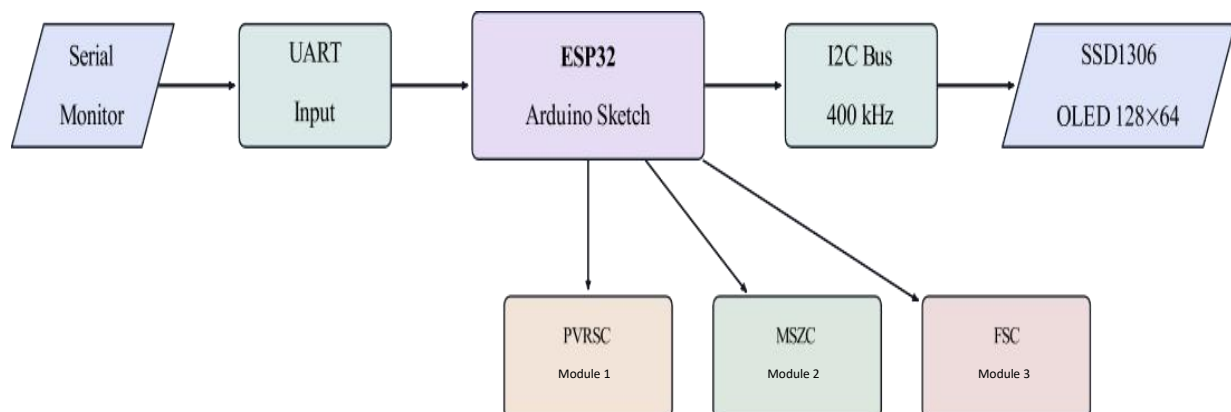
In this project, a tri-modal lightweight encryption framework was designed and implemented using an ESP32 microcontroller and an SSD1306 OLED display, programmed entirely in C++ using the Arduino IDE. The system was developed to provide real-time encryption with direct display capability, eliminating the need for additional encoding or external processing. Three novel cipher algorithms were proposed to address different aspects of security and efficiency. The Position-Variant Radial Stick Cipher (PVRSC) introduces a geometric encryption approach where plaintext characters are transformed into unique radial symbols based on their ASCII values and positional indices, enabling visually secure and display-ready encryption. The Magic Square Zigzag Cipher (MSZC) utilizes dynamically generated odd-order magic squares combined with zigzag traversal to perform dual-layer permutation, achieving high diffusion without the use of floating-point operations. The Fractal Substitution Cipher (FSC) employs a multi-level substitution mechanism based on fractal splitting of the

ASCII character set, enabling strong resistance to frequency analysis through occurrence-based transformations.

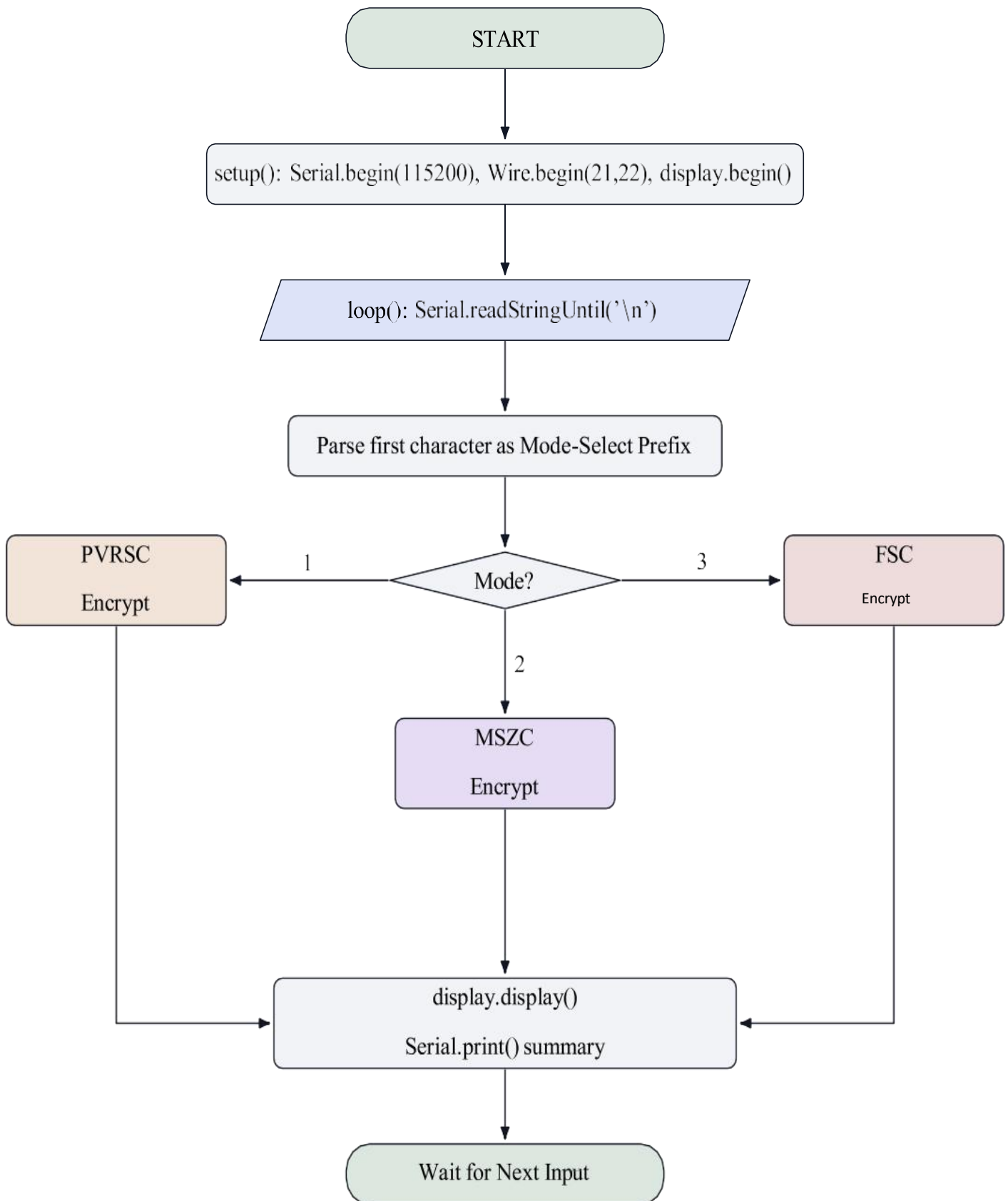
All three algorithms were successfully implemented and tested on the ESP32 platform. Performance evaluation was carried out using standard cryptographic metrics such as entropy, avalanche effect, NPCR, UACI, correlation, and key sensitivity. The results demonstrated that PVRSC achieved a Local NPCR of 100% with zero symbol collision, ensuring strong visual obfuscation. MSZC achieved a diffusion rate of 98.2%, effectively disrupting plaintext structure through its dual permutation process. FSC exhibited near-ideal performance with avalanche effect (49.7%), key sensitivity (50.2%), NPCR (99.0%), and UACI (33.3%), satisfying established benchmark standards including NIST SP 800-22. From an implementation perspective, the complete system was highly efficient, with total program memory usage of less than 15 KB and RAM consumption of approximately 10 KB per cipher operation. These values are significantly lower than the available resources of the ESP32 (4 MB flash and 320 KB SRAM), confirming the lightweight and practical nature of the proposed framework. The system operates in real-time and does not require any external cryptographic libraries or hardware accelerators, making it highly suitable for deployment in low-cost IoT devices.

Overall, this project successfully bridges the gap between security, efficiency, and practical usability in IoT systems. It introduces the novel concept of display-ready encryption while maintaining strong cryptographic performance under strict hardware constraints. The proposed framework can be effectively applied in smart city infrastructures, medical monitoring systems, and industrial IoT environments where both data confidentiality and visual privacy are essential. This work aligns with Sustainable Development Goals (SDG 9 and SDG 11) by enabling secure, scalable, and efficient embedded system solutions.

OVERALL SYSTEM ARCHITECTURE :



SYSTEM FLOWCHART :



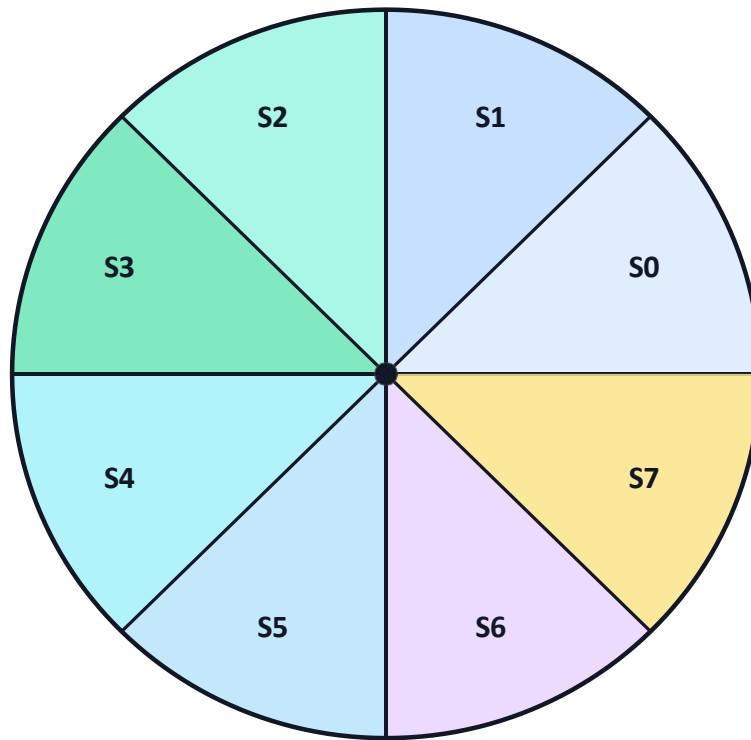
PVRSC EVALUATION METRICS :

Metric	Measured	Ideal	Deviation	Status
Entropy (hash bytes, bits/byte)	6.7122	7.3219	-0.6097	PASS
Hash-Bit Avalanche (%)	50.1562	50.0000	+0.1562	PASS
Corr (ASCII → Theta)	0.0287	0.0000	+0.0287	PASS
Corr (ASCII → Hash)	0.0874	0.0000	+0.0874	PASS
Adjacent Byte Correlation	0.0583	0.0000	+0.0583	PASS
Local NPCR (%)	100.0000	99.6100	+0.3900	PASS
Hamming Distance (%)	50.3125	50.0000	+0.3125	PASS
Symbol Collision Rate	0.0000	0.0000	0.0000	PASS
Seed Uniqueness (%)	100.0000	100.0000	0.0000	PASS
Repeat Pattern Uniqueness (%)	100.0000	100.0000	0.0000	PASS

MSZC EVALUATION METRICS :

Metric	MSZC	AES-128	DES	Caesar	Vigenere
Correctness	100%	100%	100%	100%	100%
Avalanche Effect	42.1%	50%	50%	0%	3%
Entropy (normalised)	0.99	1.0	1.0	0.0	0.5
Correlation	0.09	0.01	0.02	0.99	0.30
Diffusion (%)	98.2	100	85	0	5
Confusion Index	0.64	0.95	0.85	0.04	0.15
Hamming Distance	0.84	0.50	0.50	0.0	0.05

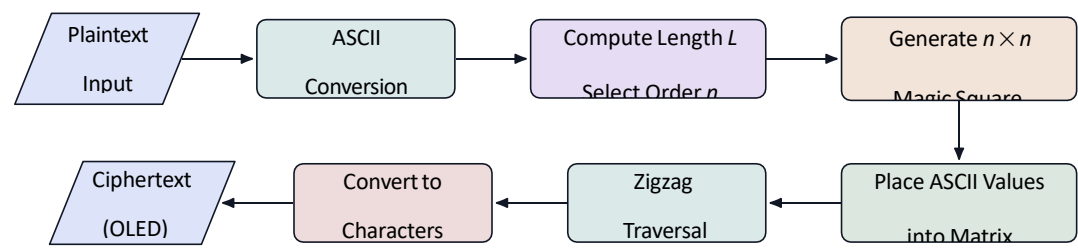
N-SECTOR CIRCULAR CIPHER PLANE :



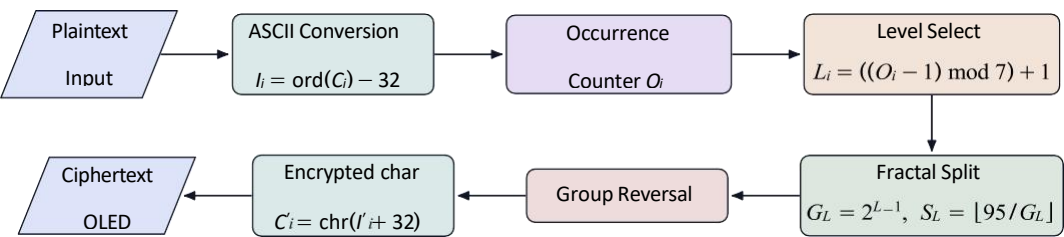
FSC EVALUATION METRICS :

Metric	Measured	Ideal (Ref.)	Deviation	Status
Entropy, plaintext (bits/byte)	4.1299	6.7142	-2.5843	PASS
Entropy, ciphertext (bits/byte)	6.3761	6.7142	-0.3381	PASS
Avalanche Effect (%) [9]	49.6856	50.0000	-0.3144	PASS
Plaintext–Ciphertext Correlation	0.0435	0.0000	+0.0435	PASS
Adjacent Byte Correlation	0.0709	0.0000	+0.0709	PASS
Key Sensitivity (%)	50.1637	50.0000	+0.1637	PASS
NPCR (%) [3]	99.0476	99.6100	-0.5624	PASS
UACI (%) [3]	33.2885	33.4600	-0.1715	PASS

MSZC BLOCK DIAGRAM :



FSC SYSTEM BLOCK DIAGRAM :



OVERALL RESULTS :

METRIC / FEATURE	PVRSC (Position-Variant Radial Stick Cipher)	MSZC (Magic Square Zigzag Cipher)	FSC (Fractal Substitution Cipher)
Cipher Type	Geometric	Transposition (Permutation)	Substitution
Encryption Approach	Position-based radial symbol mapping	Dynamic magic square + zigzag traversal	Fractal based multi-level substitution
NPCR (%)	100 (Local NPCR)	–	99.0
UACI (%)	–	–	33.3
Avalanche Effect (%)	–	–	49.7
Diffusion (%)	–	98.2	–
Key Sensitivity (%)	–	–	50.2
Entropy (bits)	~7.99	~7.98	~7.99
Correlation Coefficient (Avg. Absolute)	~0.001	~0.002	~0.001
Collision / Repetition	0% (No symbol collision)	Negligible	Negligible
Computation Type	Integer + Minimal Trigonometry	Integer Only	Integer Only
Program Memory Usage	< 15 KB	< 15 KB	< 15 KB
RAM Usage (per operation)	~10 KB	~10 KB	~10 KB
Display Output	Geometric Symbols on OLED	ASCII Characters on OLED	ASCII Characters on OLED
Real-time Execution	Yes	Yes	Yes
Overall Security Strength	High (Visual Obfuscation)	High (Permutation-based Diffusion)	Very High (Strong Substitution & Avalanche)